

PHYSICS SIMS SIUT

PHYSICS - ALL TOPICS (20 MCQs EACH)

MECHANICS (Motion, Force, Work & Energy) - 20 MCQs

1. The rate of change of displacement is called:

- A) Acceleration
- B) Velocity ✓
- C) Speed
- D) Momentum

Answer: B

2. The rate of change of velocity is called:

- A) Speed
- B) Displacement
- C) Acceleration ✓
- D) Momentum

Answer: C

3. Newton's First Law is also known as:

- A) Law of Gravitation
- B) Law of Inertia ✓
- C) Law of Action-Reaction
- D) Law of Momentum

Answer: B

4. Newton's Second Law states that force is equal to:

- A) Mass $\times$ Acceleration ✓
- B) Mass $\times$ Velocity
- C) Mass $\times$ Displacement
- D) Velocity $\times$ Time

Answer: A

5. Newton's Third Law states that:

- A) Every action has equal and opposite reaction ✓
- B) Force is mass times acceleration
- C) Energy cannot be created or destroyed
- D) The universe is expanding

Answer: A

6. The product of mass and velocity is known as:

- A) Force
- B) Acceleration
- C) Momentum ✓
- D) Work

Answer: C

7. The SI unit of force is:

- A) Joule
- B) Watt
- C) Newton ✓
- D) Pascal

Answer: C

8. Work is defined as:

- A) Force $\times$ Distance ✓
- B) Force $\times$ Time
- C) Mass $\times$ Acceleration
- D) Power $\times$ Time

Answer: A

9. The SI unit of work is:

- A) Newton
- B) Joule ✓
- C) Watt
- D) Pascal

Answer: B

10. Power is defined as:

- A) Work $\times$ Time
- B) Work / Time ✓
- C) Force $\times$ Distance
- D) Mass $\times$ Acceleration

Answer: B

11. The SI unit of power is:

- A) Joule
- B) Newton
- C) Watt ✓
- D) Pascal

Answer: C

12. Kinetic energy is given by:

- A) $\frac{1}{2} mv^2$ ✓
- B) mgh
- C) mv
- D) $\frac{1}{2} kx^2$

Answer: A

13. Potential energy is given by:

- A) $\frac{1}{2} mv^2$
- B) mgh ✓
- C) mv
- D) $\frac{1}{2} kx^2$

Answer: B

14. The law of conservation of energy states that:

- A) Energy can be created
- B) Energy can be destroyed
- C) Energy cannot be created or destroyed ✓
- D) Energy is always lost

Answer: C

15. Friction always acts in:

- A) Direction of motion
- B) Opposite direction of motion ✓
- C) Upward direction
- D) Downward direction

Answer: B

16. In a perfectly elastic collision:

- A) Only momentum is conserved
- B) Only kinetic energy is conserved
- C) Both momentum and kinetic energy are conserved ✓
- D) Neither is conserved

Answer: C

17. The force that keeps a body moving in a circular path is called:

- A) Centrifugal force
- B) Centripetal force ✓
- C) Frictional force
- D) Gravitational force

Answer: B

18. The moment of inertia depends on:

- A) Mass only
- B) Shape only
- C) Mass and distribution of mass ✓
- D) Velocity only

Answer: C

19. Angular momentum is conserved when:

- A) Net force is zero
- B) Net torque is zero ✓
- C) Net work is zero
- D) Net power is zero

Answer: B

20. The work-energy theorem states that:

- A) Work = Force × Distance
- B) Work = Change in kinetic energy ✓
- C) Work = Change in potential energy
- D) Work = Power × Time

Answer: B

GRAVITATION - 20 MCQs

1. The force of attraction between two masses is called:

- A) Magnetic force
- B) Gravitational force ✓
- C) Electrostatic force
- D) Nuclear force

Answer: B

2. Newton's law of gravitation states that $F =$:

- A) Gm_1m_2/r
- B) Gm_1m_2/r^2 ✓
- C) Gm_1m_2/r^3
- D) Gm_1m_2/r^4

Answer: B

3. The value of gravitational constant G is:

- A) $6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$ ✓
- B) $6.67 \times 10^{11} \text{ Nm}^2/\text{kg}^2$
- C) 9.8 m/s^2
- D) $6.67 \times 10^{-10} \text{ Nm}^2/\text{kg}^2$

Answer: A

4. The value of acceleration due to gravity on Earth is approximately:

- A) 6.67 m/s^2
- B) 9.8 m/s^2 ✓
- C) 10.8 m/s^2

D) 11.2 m/s^2

Answer: B

5. The value of g decreases with:

- A) Increase in altitude ✓
- B) Decrease in altitude
- C) Increase in mass
- D) None of these

Answer: A

6. The value of g is maximum at:

- A) Poles ✓
- B) Equator
- C) Center of Earth
- D) All places

Answer: A

7. The value of g is minimum at:

- A) Poles
- B) Equator ✓
- C) Center of Earth
- D) All places

Answer: B

8. The escape velocity from Earth's surface is approximately:

- A) 7.9 km/s
- B) 11.2 km/s ✓
- C) 14.7 km/s
- D) 20.5 km/s

Answer: B

9. The orbital velocity of a satellite near Earth's surface is approximately:

- A) 7.9 km/s ✓
- B) 11.2 km/s
- C) 14.7 km/s
- D) 20.5 km/s

Answer: A

10. The weight of a body is:

- A) Mass $\times$ Velocity
- B) Mass $\times$ Acceleration due to gravity ✓
- C) Mass $\times$ Distance
- D) Mass $\times$ Time

Answer: B

11. Weightlessness occurs when:

- A) $g = 0$ ✓
- B) $g = 9.8$
- C) $g = \infty$
- D) $g = \text{constant}$

Answer: A

12. The Kepler's law of periods (Third Law) states that $T^2 \propto$:

- A) r
- B) r^2
- C) r^3 ✓
- D) r^4

Answer: C

13. The gravitational force between two masses:

- A) Is always attractive ✓
- B) Is always repulsive
- C) Can be attractive or repulsive
- D) Depends on charge

Answer: A

14. The gravitational field is a:

- A) Scalar quantity
- B) Vector quantity ✓
- C) Tensor quantity
- D) None of these

Answer: B

15. The SI unit of gravitational field strength is:

- A) N
- B) N/kg ✓
- C) kg/N
- D) J/kg

Answer: B

16. The gravitational potential energy at height h is:

- A) mgh ✓
- B) $\frac{1}{2}mv^2$
- C) GMm/r
- D) Gm_1m_2/r^2

Answer: A

17. The gravitational potential at infinity is:

- A) Zero ✓
- B) Maximum
- C) Minimum
- D) Infinite

Answer: A

18. The orbital speed of a satellite depends on:

- A) Mass of satellite
- B) Mass of Earth and orbital radius ✓
- C) Size of satellite
- D) Shape of satellite

Answer: B

19. The geostationary orbit is at approximately:

- A) 100 km
- B) 500 km
- C) 36,000 km ✓
- D) 72,000 km

Answer: C

20. The time period of a geostationary satellite is:

- A) 12 hours
- B) 24 hours ✓
- C) 48 hours
- D) 72 hours

Answer: B

FLUIDS (Pressure, Buoyancy, Flow) - 20 MCQs

1. The force exerted per unit area is called:

- A) Force
- B) Pressure ✓
- C) Buoyancy
- D) Viscosity

Answer: B

2. The SI unit of pressure is:

- A) Newton
- B) Joule
- C) Pascal ✓
- D) Watt

Answer: C

3. The pressure exerted by a liquid at a depth h is:

- A) ρgh ✓
- B) $\rho g/h$
- C) gh/ρ
- D) ρ/g

Answer: A

4. The upward force experienced by a body immersed in a fluid is called:

- A) Pressure
- B) Buoyant force ✓
- C) Viscous force
- D) Frictional force

Answer: B

5. The principle that states that the buoyant force is equal to the weight of displaced fluid is:

- A) Pascal's Law
- B) Archimedes' Principle ✓
- C) Bernoulli's Principle
- D) Boyle's Law

Answer: B

6. The pressure applied to an enclosed fluid is transmitted equally to all parts of the fluid. This is:

- A) Pascal's Law ✓
- B) Archimedes' Principle
- C) Bernoulli's Principle
- D) Boyle's Law

Answer: A

7. The equation of continuity is based on the conservation of:

- A) Mass ✓
- B) Energy
- C) Momentum
- D) Pressure

Answer: A

8. Bernoulli's equation is based on the conservation of:

- A) Mass
- B) Energy ✓
- C) Momentum

D) Pressure

Answer: B

9. The property of a fluid that resists flow is called:

- A) Pressure
- B) Buoyancy
- C) Viscosity ✓
- D) Density

Answer: C

10. According to Bernoulli's equation, as the speed of a fluid increases, its pressure:

- A) Increases
- B) Decreases ✓
- C) Remains constant
- D) Becomes zero

Answer: B

11. The density of water is approximately:

- A) 1 g/cm³ ✓
- B) 10 g/cm³
- C) 0.1 g/cm³
- D) 100 g/cm³

Answer: A

12. An object will float if its density is:

- A) Greater than the fluid
- B) Less than the fluid ✓
- C) Equal to the fluid
- D) None of these

Answer: B

13. An object will sink if its density is:

- A) Greater than the fluid ✓
- B) Less than the fluid
- C) Equal to the fluid
- D) None of these

Answer: A

14. The pressure at a point in a fluid at rest is:

- A) Same in all directions ✓
- B) Different in different directions
- C) Zero
- D) Maximum at the surface

Answer: A

15. The flow of a fluid where the velocity at any point is constant is called:

- A) Turbulent flow
- B) Steady flow ✓
- C) Viscous flow
- D) None of these

Answer: B

16. The flow of a fluid where the velocity varies randomly is called:

- A) Steady flow
- B) Turbulent flow ✓
- C) Laminar flow
- D) None of these

Answer: B

17. The ratio of shear stress to shear strain rate is called:

- A) Pressure
- B) Density
- C) Viscosity ✓
- D) Buoyancy

Answer: C

18. The surface of a liquid behaves like a stretched membrane due to:

- A) Viscosity
- B) Surface tension ✓
- C) Buoyancy
- D) Pressure

Answer: B

19. The rise of a liquid in a narrow tube is due to:

- A) Viscosity
- B) Surface tension ✓
- C) Buoyancy
- D) Pressure

Answer: B

20. The SI unit of viscosity is:

- A) $\text{Pa} \cdot \text{s}$ ✓
- B) N/m^2
- C) J/s
- D) kg/m^3

Answer: A

OSCILLATIONS & SIMPLE HARMONIC MOTION - 20 MCQs

1. The to-and-fro motion of a body about a mean position is called:

- A) Circular motion
- B) Oscillatory motion ✓
- C) Linear motion
- D) Rotational motion

Answer: B

2. Simple Harmonic Motion is characterized by acceleration proportional to:

- A) Velocity
- B) Displacement ✓
- C) Time
- D) Frequency

Answer: B

3. In SHM, the acceleration is always directed towards:

- A) Mean position ✓
- B) Extreme position
- C) Any position
- D) None of these

Answer: A

4. The SI unit of frequency is:

- A) Hertz ✓

- B) Joule
- C) Newton
- D) Pascal

Answer: A

5. The time period of a pendulum depends on:

- A) Mass of bob
- B) Length of pendulum ✓
- C) Amplitude
- D) None of these

Answer: B

6. The time period of a mass-spring system depends on:

- A) Mass and spring constant ✓
- B) Amplitude
- C) Frequency
- D) None of these

Answer: A

7. The maximum displacement from the mean position is called:

- A) Frequency
- B) Amplitude ✓
- C) Period
- D) Phase

Answer: B

8. The number of oscillations per unit time is called:

- A) Amplitude

- B) Frequency ✓
- C) Period
- D) Phase

Answer: B

9. The time taken for one complete oscillation is called:

- A) Amplitude
- B) Frequency
- C) Period ✓
- D) Phase

Answer: C

10. The relationship between frequency (f) and time period (T) is:

- A) $f = T$
- B) $f = 1/T$ ✓
- C) $f = T^2$
- D) $f = 1/T^2$

Answer: B

11. The energy of a simple harmonic oscillator is:

- A) Constant ✓
- B) Increasing
- C) Decreasing
- D) Zero

Answer: A

12. The maximum velocity in SHM occurs at:

- A) Mean position ✓

- B) Extreme position
- C) Mid-point
- D) None of these

Answer: A

13. The maximum acceleration in SHM occurs at:

- A) Mean position
- B) Extreme position ✓
- C) Mid-point
- D) None of these

Answer: B

14. The phase of a particle in SHM describes its:

- A) Position and direction of motion ✓
- B) Only position
- C) Only velocity
- D) Only acceleration

Answer: A

15. A simple pendulum performs:

- A) Linear motion
- B) Circular motion
- C) Simple harmonic motion ✓
- D) Rotational motion

Answer: C

16. The restoring force in SHM is directly proportional to:

- A) Velocity
- B) Displacement ✓

- C) Time
- D) Frequency

Answer: B

17. The oscillation of a pendulum is isochronous. This means:

- A) Time period depends on amplitude
- B) Time period is independent of amplitude for small amplitudes ✓
- C) Time period depends on mass
- D) None of these

Answer: B

18. The graph of displacement vs time for SHM is:

- A) Straight line
- B) Parabola
- C) Sine/Cosine wave ✓
- D) Circle

Answer: C

19. Damping in an oscillatory system causes the amplitude to:

- A) Increase
- B) Decrease ✓
- C) Remain constant
- D) Become zero instantly

Answer: B

20. Resonance occurs when the driving frequency is equal to the:

- A) Natural frequency ✓

- B) Amplitude
- C) Time period
- D) None of these

Answer: A

WAVES & SOUND - 20 MCQs

1. The propagation of disturbance through a medium is called:

- A) Wave ✓
- B) Particle
- C) Force
- D) Energy

Answer: A

2. Waves that require a medium to propagate are called:

- A) Mechanical waves ✓
- B) Electromagnetic waves
- C) Matter waves
- D) None of these

Answer: A

3. Waves that do not require a medium to propagate are called:

- A) Mechanical waves
- B) Electromagnetic waves ✓
- C) Matter waves
- D) None of these

Answer: B

4. The speed of a wave is given by:

- A) $f\lambda$ ✓
- B) f/λ
- C) λ/f
- D) $f \times A$

Answer: A

5. The distance between two consecutive crests or troughs is called:

- A) Amplitude
- B) Frequency
- C) Wavelength ✓
- D) Period

Answer: C

6. The maximum displacement of a wave from its mean position is called:

- A) Amplitude ✓
- B) Frequency
- C) Wavelength
- D) Period

Answer: A

7. The number of waves passing a point per second is called:

- A) Amplitude
- B) Frequency ✓
- C) Wavelength
- D) Period

Answer: B

8. The relationship between speed (v), frequency (f) and wavelength (λ) is:

- A) $v = f/\lambda$
- B) $v = f\lambda$ ✓
- C) $v = \lambda/f$
- D) $v = f + \lambda$

Answer: B

9. Longitudinal waves consist of:

- A) Crests and troughs
- B) Compressions and rarefactions ✓
- C) Only crests
- D) Only troughs

Answer: B

10. Transverse waves consist of:

- A) Compressions and rarefactions
- B) Crests and troughs ✓
- C) Only compressions
- D) Only rarefactions

Answer: B

11. The SI unit of intensity of sound is:

- A) Hertz
- B) Joule
- C) Watt/m^2 ✓
- D) Newton

Answer: C

12. The loudness of sound depends on:

- A) Frequency
- B) Amplitude/Intensity ✓
- C) Speed
- D) Wavelength

Answer: B

13. The pitch of sound depends on:

- A) Amplitude
- B) Frequency ✓
- C) Speed
- D) Wavelength

Answer: B

14. The unit of frequency is:

- A) Hertz ✓
- B) Joule
- C) Newton
- D) Pascal

Answer: A

15. The speed of sound is maximum in:

- A) Gases
- B) Liquids
- C) Solids ✓
- D) Vacuum

Answer: C

16. The speed of sound is minimum in:

- A) Gases ✓
- B) Liquids
- C) Solids
- D) None of these

Answer: A

17. The speed of sound in air is approximately:

- A) 340 m/s ✓
- B) 500 m/s
- C) 1000 m/s
- D) 3000 m/s

Answer: A

18. The phenomenon of two waves overlapping to produce a resultant wave is called:

- A) Reflection
- B) Refraction
- C) Interference ✓
- D) Diffraction

Answer: C

19. Beats are formed due to:

- A) Reflection
- B) Interference ✓
- C) Refraction
- D) Diffraction

Answer: B

20. The Doppler effect is the change in frequency due to:

- A) Motion of source/observer ✓
- B) Change in medium
- C) Change in temperature
- D) None of these

Answer: A

HEAT & THERMODYNAMICS - 20 MCQs

1. The measure of the average kinetic energy of the particles of a substance is called:

- A) Heat
- B) Temperature ✓
- C) Internal energy
- D) Entropy

Answer: B

2. The SI unit of heat is:

- A) Joule ✓
- B) Newton
- C) Pascal
- D) Watt

Answer: A

3. The temperature at which all molecular motion stops is called:

- A) Absolute zero ✓
- B) Critical temperature
- C) Boiling point
- D) Melting point

Answer: A

4. Absolute zero is:

- A) 0°C
- B) 0 K ✓
- C) -273°C
- D) Both B and C

Answer: D

5. The amount of heat required to raise the temperature of 1 kg of a substance by 1 K is called:

- A) Latent heat
- B) Specific heat ✓
- C) Heat capacity
- D) Thermal conductivity

Answer: B

6. The specific heat of water is approximately:

- A) $4200\text{ J/kg}\cdot\text{K}$ ✓
- B) $2000\text{ J/kg}\cdot\text{K}$
- C) $1000\text{ J/kg}\cdot\text{K}$
- D) $800\text{ J/kg}\cdot\text{K}$

Answer: A

7. The amount of heat required to change the state of a substance at constant temperature is called:

- A) Specific heat
- B) Heat capacity
- C) Latent heat ✓
- D) Thermal energy

Answer: C

8. The boiling point of water at standard pressure is:

- A) 0°C
- B) 100°C ✓
- C) 50°C
- D) 200°C

Answer: B

9. The melting point of ice at standard pressure is:

- A) -10°C
- B) 0°C ✓
- C) 10°C
- D) 20°C

Answer: B

10. A process in which no heat is exchanged with the surroundings is called:

- A) Isothermal
- B) Isobaric
- C) Adiabatic ✓
- D) Isochoric

Answer: C

11. A process in which temperature remains constant is called:

- A) Isothermal ✓
- B) Isobaric
- C) Adiabatic
- D) Isochoric

Answer: A

12. A process in which pressure remains constant is called:

- A) Isothermal
- B) Isobaric ✓
- C) Adiabatic
- D) Isochoric

Answer: B

13. A process in which volume remains constant is called:

- A) Isothermal
- B) Isobaric
- C) Adiabatic
- D) Isochoric ✓

Answer: D

14. The first law of thermodynamics is based on the law of conservation of:

- A) Mass
- B) Energy ✓
- C) Momentum
- D) Charge

Answer: B

15. The first law of thermodynamics is expressed as:

- A) $\Delta U = Q + W$
- B) $\Delta U = Q - W$ ✓
- C) $\Delta U = W - Q$
- D) $\Delta U = Q \times W$

Answer: B

16. The second law of thermodynamics is concerned with:

- A) Conservation of energy
- B) Direction of heat flow ✓
- C) Conservation of mass
- D) None of these

Answer: B

17. A Carnot engine is a:

- A) Real engine
- B) Ideal engine ✓
- C) Refrigerator
- D) Heat pump

Answer: B

18. The efficiency of a Carnot engine depends on:

- A) Temperature of source and sink ✓
- B) Working substance
- C) Size of engine
- D) None of these

Answer: A

19. Entropy is a measure of:

- A) Energy
- B) Disorder ✓
- C) Temperature
- D) Pressure

Answer: B

20. Heat transfer through the actual movement of fluid molecules is called:

- A) Conduction
- B) Convection ✓
- C) Radiation
- D) Absorption

Answer: B

LIGHT & OPTICS (Reflection, Refraction, Lenses) - 20 MCQs

1. The branch of physics dealing with light is called:

- A) Optics ✓
- B) Electricity
- C) Magnetism
- D) Mechanics

Answer: A

2. The speed of light in vacuum is approximately:

- A) 3×10^8 m/s ✓
- B) 3×10^6 m/s
- C) 3×10^{10} m/s
- D) 3×10^4 m/s

Answer: A

3. The bouncing back of light from a surface is called:

- A) Refraction
- B) Reflection ✓

- C) Diffraction
- D) Interference

Answer: B

4. The bending of light when it passes from one medium to another is called:

- A) Reflection
- B) Refraction ✓
- C) Diffraction
- D) Interference

Answer: B

5. The law of reflection states that the angle of incidence is:

- A) Equal to the angle of reflection ✓
- B) Greater than the angle of reflection
- C) Less than the angle of reflection
- D) None of these

Answer: A

6. The refractive index (n) is given by:

- A) Speed of light in vacuum / Speed of light in medium ✓
- B) Speed of light in medium / Speed of light in vacuum
- C) Speed of light in air / Speed of light in vacuum
- D) None of these

Answer: A

7. Snell's law relates:

- A) Reflection and refraction

- B) Angle of incidence and angle of refraction ✓
C) Speed of light and wavelength
D) None of these

Answer: B

8. Snell's law is expressed as:

- A) $n_1 \sin\theta_1 = n_2 \sin\theta_2$ ✓
B) $n_1 \cos\theta_1 = n_2 \cos\theta_2$
C) $n_1 \tan\theta_1 = n_2 \tan\theta_2$
D) None of these

Answer: A

9. A convex lens is also known as:

- A) Converging lens ✓
B) Diverging lens
C) Concave lens
D) Cylindrical lens

Answer: A

10. A concave lens is also known as:

- A) Converging lens
B) Diverging lens ✓
C) Convex lens
D) Cylindrical lens

Answer: B

11. The focal length of a convex lens is:

- A) Positive ✓
B) Negative
C) Zero
D) Infinite

Answer: A

12. The focal length of a concave lens is:

- A) Positive
B) Negative ✓
C) Zero
D) Infinite

Answer: B

13. The power of a lens is measured in:

- A) Diopters ✓
B) Watts
C) Joules
D) Newtons

Answer: A

14. The power of a lens is given by:

- A) $P = 1/f$ (in meters) ✓
B) $P = f$
C) $P = 1/f^2$
D) $P = f^2$

Answer: A

15. A real image is:

- A) Formed on a screen ✓
B) Cannot be formed on a screen
C) Virtual
D) None of these

Answer: A

16. A virtual image is:

- A) Formed on a screen
- B) Cannot be formed on a screen ✓
- C) Real
- D) None of these

Answer: B

17. The image formed by a concave mirror is:

- A) Always real
- B) Always virtual
- C) Can be real or virtual ✓
- D) None of these

Answer: C

18. The image formed by a convex mirror is:

- A) Always real
- B) Always virtual ✓
- C) Can be real or virtual
- D) None of these

Answer: B

19. The phenomenon of splitting of white light into its component colors is called:

- A) Refraction
- B) Reflection
- C) Dispersion ✓
- D) Diffraction

Answer: C

20. The rainbow is formed due to:

- A) Reflection only
- B) Refraction only
- C) Dispersion and total internal reflection ✓
- D) Diffraction

Answer: C

ELECTROSTATICS - 20 MCQs

1. The study of charges at rest is called:

- A) Electrostatics ✓
- B) Current electricity
- C) Magnetism
- D) Electromagnetism

Answer: A

2. Like charges:

- A) Attract each other
- B) Repel each other ✓
- C) Neither attract nor repel
- D) None of these

Answer: B

3. Unlike charges:

- A) Attract each other ✓
- B) Repel each other
- C) Neither attract nor repel
- D) None of these

Answer: A

4. The SI unit of charge is:

- A) Coulomb ✓
- B) Ampere
- C) Volt
- D) Ohm

Answer: A

5. The charge on an electron is:

- A) $-1.6 \times 10^{-19} \text{ C}$ ✓
- B) $+1.6 \times 10^{-19} \text{ C}$
- C) $-1.6 \times 10^{19} \text{ C}$
- D) $+1.6 \times 10^{19} \text{ C}$

Answer: A

6. The charge on a proton is:

- A) $-1.6 \times 10^{-19} \text{ C}$
- B) $+1.6 \times 10^{-19} \text{ C}$ ✓
- C) $-1.6 \times 10^{19} \text{ C}$
- D) $+1.6 \times 10^{19} \text{ C}$

Answer: B

7. Coulomb's law states that the force between two charges is:

- A) Directly proportional to the square of the distance
- B) Inversely proportional to the square of the distance ✓
- C) Directly proportional to the distance
- D) Inversely proportional to the distance

Answer: B

8. The electric field is defined as:

- A) Force / Charge ✓
- B) Charge / Force
- C) Force $\times$ Charge
- D) None of these

Answer: A

9. The SI unit of electric field is:

- A) N/C ✓
- B) C/N
- C) V/m
- D) Both A and C

Answer: D

10. The electric potential is defined as:

- A) Work / Charge ✓
- B) Charge / Work
- C) Work $\times$ Charge
- D) None of these

Answer: A

11. The SI unit of electric potential is:

- A) Joule
- B) Volt ✓
- C) Coulomb
- D) Newton

Answer: B

12. Capacitance is defined as:

- A) Charge / Voltage ✓
- B) Voltage / Charge
- C) Charge $\times$ Voltage

D) None of these

Answer: A

13. The SI unit of capacitance is:

- A) Farad ✓
- B) Coulomb
- C) Volt
- D) Ohm

Answer: A

14. A capacitor is a device that stores:

- A) Current
- B) Charge ✓
- C) Resistance
- D) Power

Answer: B

15. The capacitance of a parallel plate capacitor depends on:

- A) Area of plates
- B) Distance between plates
- C) Medium between plates
- D) All of these ✓

Answer: D

16. A dielectric is a material that:

- A) Conducts electricity
- B) Insulates electricity ✓
- C) Creates charge
- D) None of these

Answer: B

17. The electric field inside a conductor is:

- A) Maximum
- B) Minimum
- C) Zero ✓
- D) None of these

Answer: C

18. The electric field lines:

- A) Start from negative charges
- B) Start from positive charges ✓
- C) Start from both charges
- D) None of these

Answer: B

19. The electric field lines end at:

- A) Positive charges
- B) Negative charges ✓
- C) Both charges
- D) None of these

Answer: B

20. The electric potential at infinity is:

- A) Maximum
- B) Minimum
- C) Zero ✓
- D) Infinite

Answer: C

CURRENT ELECTRICITY & CIRCUITS - 20 MCQs

1. The flow of electric charge is called:

- A) Voltage
- B) Current ✓
- C) Resistance
- D) Power

Answer: B

2. The SI unit of current is:

- A) Volt
- B) Ampere ✓
- C) Ohm
- D) Watt

Answer: B

3. The potential difference across a conductor is measured in:

- A) Ampere
- B) Volt ✓
- C) Ohm
- D) Watt

Answer: B

4. The resistance of a conductor is measured in:

- A) Ampere
- B) Volt
- C) Ohm ✓
- D) Watt

Answer: C

5. Ohm's law states that $V =$:

- A) IR ✓
- B) I/R
- C) R/I
- D) I^2R

Answer: A

6. The power dissipated in a resistor is given by:

- A) $P = IV$ ✓
- B) $P = I/V$
- C) $P = V/I$
- D) $P = I^2V$

Answer: A

7. The resistance of a conductor depends on:

- A) Length
- B) Cross-sectional area
- C) Material
- D) All of these ✓

Answer: D

8. In a series circuit, the current is:

- A) Different in each resistor
- B) The same in all resistors ✓
- C) Zero
- D) None of these

Answer: B

9. In a parallel circuit, the voltage across each branch is:

- A) Different
- B) The same ✓
- C) Zero
- D) None of these

Answer: B

10. The equivalent resistance in series is:

- A) $R_1 + R_2 + R_3 + \dots$ ✓
- B) $1/R_1 + 1/R_2 + 1/R_3 + \dots$
- C) $R_1 \times R_2 \times R_3$
- D) None of these

Answer: A

11. The equivalent resistance in parallel is given by:

- A) $R_1 + R_2 + R_3$
- B) $1/R_1 + 1/R_2 + 1/R_3$ ✓
- C) $1/(R_1 + R_2 + R_3)$
- D) $R_1 \times R_2 \times R_3$

Answer: B

12. A short circuit has:

- A) Very high resistance
- B) Very low resistance ✓
- C) No resistance
- D) None of these

Answer: B

13. An open circuit has:

- A) Very high resistance ✓
- B) Very low resistance
- C) No resistance
- D) None of these

Answer: A

14. The SI unit of power is:

- A) Joule
- B) Watt ✓
- C) Newton
- D) Pascal

Answer: B

15. Electric energy is measured in:

- A) Joule ✓
- B) Watt
- C) Newton
- D) Pascal

Answer: A

16. Kilowatt-hour is a unit of:

- A) Power
- B) Energy ✓
- C) Current
- D) Voltage

Answer: B

17. The filament of a bulb is made of:

- A) Copper
- B) Tungsten ✓
- C) Iron

D) Aluminum

Answer: B

18. A fuse wire has:

- A) High resistance
- B) Low resistance
- C) High melting point
- D) Low melting point ✓

Answer: D

19. The flow of electrons in a circuit is from:

- A) Positive to negative terminal
- B) Negative to positive terminal ✓
- C) Both directions
- D) None of these

Answer: B

20. The conventional current flows from:

- A) Positive to negative terminal ✓
- B) Negative to positive terminal
- C) Both directions
- D) None of these

Answer: A

MAGNETISM & ELECTROMAGNETISM - 20 MCQs

1. The property of a material to attract iron, cobalt, and nickel is called:

- A) Electricity
- B) Magnetism ✓
- C) Gravity
- D) Friction

Answer: B

2. A magnet has two poles called:

- A) North and South ✓
- B) East and West
- C) Positive and Negative
- D) Up and Down

Answer: A

3. Like poles of a magnet:

- A) Attract each other
- B) Repel each other ✓
- C) Neither attract nor repel
- D) None of these

Answer: B

4. Unlike poles of a magnet:

- A) Attract each other ✓
- B) Repel each other
- C) Neither attract nor repel
- D) None of these

Answer: A

5. The magnetic field lines of a bar magnet:

- A) Start from the South pole
- B) Start from the North pole ✓

- C) Start from both poles
- D) None of these

Answer: B

6. The magnetic field lines of a bar magnet end at:

- A) North pole
- B) South pole ✓
- C) Both poles
- D) None of these

Answer: B

7. The Earth acts as a:

- A) Electric dipole
- B) Magnetic dipole ✓
- C) Gravitational dipole
- D) None of these

Answer: B

8. The region around a magnet where its influence can be felt is called:

- A) Electric field
- B) Magnetic field ✓
- C) Gravitational field
- D) None of these

Answer: B

9. The SI unit of magnetic field is:

- A) Tesla ✓
- B) Weber
- C) Newton
- D) Coulomb

Answer: A

10. The SI unit of magnetic flux is:

- A) Tesla
- B) Weber ✓
- C) Newton
- D) Coulomb

Answer: B

11. Oersted discovered that:

- A) Electric current produces magnetic field ✓
- B) Magnetic field produces electric current
- C) Magnetism and electricity are independent
- D) None of these

Answer: A

12. A current-carrying conductor experiences a force in a:

- A) Electric field
- B) Magnetic field ✓
- C) Gravitational field
- D) None of these

Answer: B

13. The direction of force on a current-carrying conductor in a magnetic field is given by:

- A) Right-hand rule ✓
- B) Left-hand rule
- C) Fleming's right-hand rule

D) None of these

Answer: A

14. A transformer works on the principle of:

- A) Electrostatic induction
- B) Electromagnetic induction ✓
- C) Magnetic induction
- D) None of these

Answer: B

15. The core of a transformer is made of:

- A) Copper
- B) Iron ✓
- C) Aluminum
- D) Plastic

Answer: B

16. A step-up transformer:

- A) Increases current
- B) Increases voltage ✓
- C) Decreases voltage
- D) None of these

Answer: B

17. A step-down transformer:

- A) Increases voltage
- B) Decreases voltage ✓
- C) Increases current
- D) None of these

Answer: B

18. An electric motor converts:

- A) Mechanical energy to electrical energy
- B) Electrical energy to mechanical energy ✓
- C) Heat energy to electrical energy
- D) None of these

Answer: B

19. A generator converts:

- A) Mechanical energy to electrical energy ✓
- B) Electrical energy to mechanical energy
- C) Heat energy to electrical energy
- D) None of these

Answer: A

20. Lenz's law states that the induced current opposes the:

- A) Change in magnetic flux ✓
- B) Change in electric field
- C) Change in temperature
- D) None of these

Answer: A